Pseudocode for Case Studies

Case Study 1: Attendance Recovery Computation

Pseudocode:

START

READ attended

READ conducted

DISPLAY "required 75% attendance"

DISPLAY (100 * attended / conducted)

SET to_attend = 0

SET present = 100 * attended // conducted

IF present < 75 THEN

WHILE present < 75 DO

SET to_attend = to_attend + 1

SET present = ((to_attend + attended) / (conducted + to_attend)) * 100

END WHILE

DISPLAY "you need to attend " + to_attend + " upcoming classes to acquire required attendance"

ELSE

DISPLAY "you have required attendance"

END IF

END

Case Study 2: Hospital Billing and Patient Classification

Pseudocode:

START

```
SET patients = []
READ n
FOR i FROM 0 TO n - 1 DO
DISPLAY "Enter details for Patient", i + 1
READ name
READ consultation
READ diagnostic
READ treatment
SET total = consultation + diagnostic + treatment
IF total >= 10000 THEN
SET discount = (total / 100) * 20
ELSE IF total >= 5000 THEN
SET discount = (total / 100) * 10
ELSE
SET discount = 0
END IF
SET total_bill = total - discount
IF total_bill >= 10000 THEN
SET category = "High Billing"
ELSE IF total_bill >= 5000 THEN
SET category = "Medium Billing"
ELSE
SET category = "Low Billing"
END IF
APPEND [name, total, discount, total_bill, category] TO patients
END FOR
FOR EACH patient IN patients DO
DISPLAY "Patient Name:", patient[0]
```

DISPLAY "Total Charges: Rs.", patient[1]

DISPLAY "Concession: Rs.", patient[2]

DISPLAY "Final Bill: Rs.", patient[3]

DISPLAY "Classification:", patient[4]

END FOR

DISPLAY "Total patients processed:", LENGTH(patients)

END

Case Study 3: Library Borrowing and Fine Reconciliation

Pseudocode:

START

SET valid_books = ["book1", "book2", "book3", "book4", "book5"]

SET books = ["book1"*3, "book2"*3, "book3"*3, "book4"*3, "book5"*3]

DISPLAY "available actions: take , return"

READ request

IF request == "take" THEN

READ name

IF name IN books THEN

REMOVE name FROM books

DISPLAY "return within 5 days!"

ELSE

DISPLAY "this is not available"

END IF

ELSE IF request == "return" THEN

READ name

IF name IN valid_books THEN

APPEND name TO books

```
END IF
READ time
IF time > 5 THEN
SET fine = 50
FOR i FROM 5 TO time - 1 DO
SET fine = fine + 50
END FOR
ELSE
SET fine = 0
END IF
DISPLAY "you need to pay " + fine
END IF
END
```

Case Study 4: Multi-Product Shopping Budget Constraint

Pseudocode:

```
START
SET products = []
SET total = 0
READ budget
READ n
FOR i FROM 0 TO n - 1 DO
READ name
READ price
READ quantity
SET cost = price * quantity
```

SET total = total + cost

APPEND [name, price, quantity, cost] TO products

END FOR

FOR EACH item IN products DO

DISPLAY item[0], "- Rs.", item[1], "x", item[2], "= Rs.", item[3]

END FOR

DISPLAY "Total bill: Rs.", total

IF total <= budget THEN

DISPLAY "Shopping is within your budget."

ELSE

DISPLAY "Shopping exceeds your budget."

DISPLAY "Amount exceeded: ₹", total - budget

END IF

END

Case Study 5: Multi-Stage Scholarship Shortlisting

Pseudocode:

START

SET details = []

READ name AND APPEND TO details

READ marks AND APPEND TO details

READ attendance AND APPEND TO details

READ income AND APPEND TO details

IF details[1] > 60 AND details[2] > 70 THEN

IF details[3] > 10000 THEN

SET category = 'Merit-cum-Means scholarship'

```
ELSE
SET category = 'Merit-Based scholarship'
END IF
ELSE
IF details[3] > 10000 THEN
SET category = 'Need-Based scholarship'
ELSE
SET category = 'no category'
END IF
END IF
DISPLAY "the student can apply for scholarship under " + category
END
```

Case Study 6: Password Policy Validation Engine

Pseudocode:

```
START
SET password1 = 0
WHILE password1 != "strong" DO
SET up = 0, low = 0, spe = 0, num = 0, length = 0
READ password
FOR EACH char IN password DO
IF ASCII(char) >= 48 AND ASCII(char) <= 57 THEN
SET num = num + 1
ELSE IF ASCII(char) >= 65 AND ASCII(char) <= 90 THEN
SET up = up + 1
ELSE IF ASCII(char) >= 97 AND ASCII(char) <= 122 THEN
```

```
SET low = low + 1
ELSE
SET spe = spe + 1
END IF
SET length = length + 1
END FOR
IF length > 8 THEN
IF length < 20 THEN
IF num > 0 THEN
IF low > 0 THEN
IF up > 0 THEN
IF spe > 0 THEN
DISPLAY "your password is strong"
SET password1 = "strong"
ELSE
DISPLAY "password doesn't have special characters"
SET password1 = "not strong"
END IF
ELSE
DISPLAY "password doesn't have upper characters"
SET password1 = "not strong"
END IF
ELSE
DISPLAY "password doesn't have lower characters"
SET password1 = "not strong"
END IF
ELSE
DISPLAY "password doesn't have numbers"
```

SET password1 = "not strong"

END IF

ELSE

DISPLAY "password id long"

SET password1 = "not strong"

END IF

ELSE

DISPLAY "password is short"

SET password1 = "not strong"

END IF

END WHILE

END

Case Study 7: Progressive Electricity Billing

Pseudocode:

START

READ n

SET bills = []

FOR i FROM 0 TO n - 1 DO

DISPLAY "Household", i + 1

READ name

READ units

IF units <= 100 THEN

SET bill = units * 2

ELSE IF units <= 200 THEN

SET bill = units * 3

ELSE IF units <= 300 THEN

```
SET bill = units * 5
ELSE
SET bill = units * 7
END IF
SET fixed_charge = 50
SET total_bill = bill + fixed_charge
APPEND [name, units, total_bill] TO bills
END FOR
FOR EACH h IN bills DO
DISPLAY "Household:", h[0]
DISPLAY "Units consumed:", h[1]
DISPLAY "Energy charge: Rs.", h[2] - 50
DISPLAY "Fixed charge: Rs. 50"
DISPLAY "Total bill: Rs.", h[2]
END FOR
SET highest = bills[0]
FOR EACH h IN bills DO
IF h[2] > highest[2] THEN
SET highest = h
END IF
END FOR
DISPLAY "Highest bill:", highest[0]
DISPLAY "Amount: Rs.", highest[2]
END
```

Case Study 8: Smart Canteen Order and Stock Management

Pseudocode:

START

SET rice = 20, eggs = 30, noodles = 10, chicken = 2000

SET items = ['fried rice', 'noodles', 'chicken rice']

SET special = ['eggs', 'chicken']

DISPLAY "available items are: "

FOR EACH item IN items DO

DISPLAY item

END FOR

DISPLAY "additional:"

FOR EACH s IN special DO

DISPLAY s

END FOR

SET nxt_order = 1

WHILE nxt_order != "no" DO

IF rice < 1 OR eggs < 1 OR noodles < 1 OR chicken < 200 THEN

SET nxt_order = 0

DISPLAY "out of stock"

BREAK

END IF

READ order TO LOWERCASE

READ extra TO LOWERCASE

IF order == "fried rice" THEN

SET rice = rice - 1

IF extra == "egg" THEN

SET eggs = eggs - 1

ELSE IF extra == "chicken" THEN

SET chicken = chicken - 200

END IF

ELSE IF order == "chicken rice" THEN

SET rice = rice - 1

IF extra == "egg" THEN

SET eggs = eggs - 1

END IF

ELSE IF order == "noodles" THEN

SET noodles = noodles - 1

IF extra == "egg" THEN

SET eggs = eggs - 1

ELSE IF extra == "chicken" THEN

SET chicken = chicken - 200

END IF

ELSE

DISPLAY "there no such type of item"

END IF

READ nxt_order

END WHILE

END

Case Study 9: Smart Parking Fee Management

Pseudocode:

START

SET A = ["car", "bike", "truck", "bicycle"]

DISPLAY "available slots for vehicles", A

READ parking_vehicle TO LOWERCASE AND STRIPPED

```
READ time
IF parking_vehicle == 'car' THEN
SET maximum = 10
SET fee = 100
ELSE IF parking_vehicle == 'bike' THEN
SET maximum = 5
SET fee = 50
ELSE IF parking_vehicle == 'truck' THEN
SET maximum = 15
SET fee = 200
ELSE
SET maximum = 0
SET fee = 0
END IF
IF time > maximum THEN
DISPLAY "you need to pay extra 1000"
ELSE
FOR i FROM 0 TO time - 1 DO
SET fee = fee + 20 * (fee / 100)
END FOR
END IF
DISPLAY "your total fee is", INTEGER(fee)
END
```

Case Study 10: University Scholarship Allocation

Pseudocode:

```
START
```

```
SET students = []
SET marks = []
SET attendance = []
SET income = []
READ n
FOR i FROM 0 TO n - 1 DO
READ name AND APPEND TO students
READ mark_val AND APPEND TO marks
READ att_val AND APPEND TO attendance
READ inc_val AND APPEND TO income
END FOR
SET eligible = []
FOR EACH score IN marks DO
IF score > 60 THEN
SET idx = INDEX_OF(score, marks)
APPEND students[idx] TO eligible
END IF
END FOR
DISPLAY "The eligible students are: "
FOR EACH student IN eligible DO
DISPLAY student WITHOUT NEWLINE
SET idx = INDEX_OF(student, students)
IF income[idx] < 80000 THEN
DISPLAY "with scholarship of 70%"
ELSE IF income[idx] > 120000 THEN
DISPLAY "with scholarship of 50%"
ELSE
DISPLAY "with scholarship of 30%"
```

```
END IF

END FOR

END
```